INCH-POUND

MIL-T-27/100A
13 March 1991

SUPERSEDING
MIL-T-27/100
17 MAY 1976

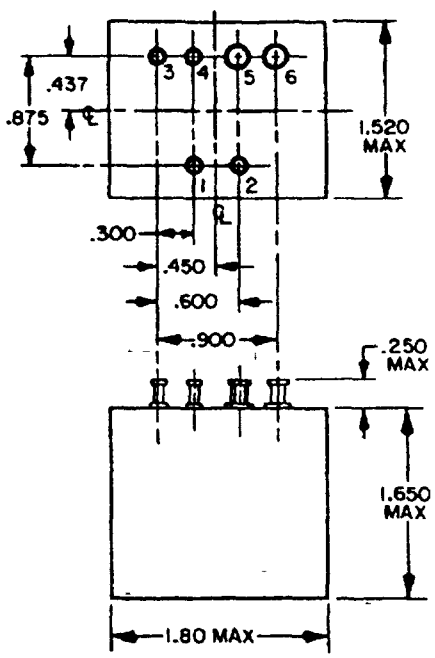
MILITARY SPECIFICATION SHEET

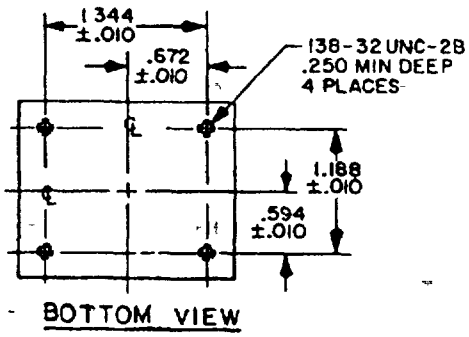
TRANSFORMER, POWER, STEPDOWN, TF5V03ZZ919

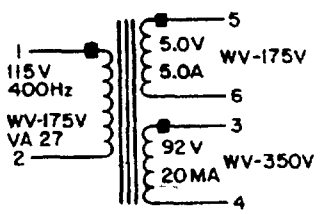
Ⓐ Inactive for new design after the date of this specification.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.







CIRCUIT DIAGRAM AND MARKING

Inches	mm
.010	0.25
.138	3.50
.250	6.35
.300	7.62
.437	11.10
.450	11.43
.594	15.09
.600	15.24
.672	17.07
.875	22.22
.900	22.86
1.188	30.18
1.344	34.14
1.520	38.61
1.650	41.91
1.80	45.72

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.030 (0.76 mm).
4. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configurations

Ⓐ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings.

Primary voltage: (1-2): 115 volts rms, 400 Hz ±5 percent, 27 voltamperes

Secondary voltage:

(3-4): 92.0 volts rms, 20 milliamperes rms.
(5-6): 5.0 volts rms, 5.0 amperes rms.

Working voltage.

(1-2): 175 volts maximum.
(3-4): 350 volts maximum.
(5-6): 175 volts maximum.

Design and construction.

Dimensions and configuration: See figure 1.

Duty cycle. Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Brass, turret type.

Terminal height. 0.250 inch maximum.

Weight: 10.0 ounces maximum.

Operating temperature range: -55°C to +155°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5.0 pounds.

Dielectric withstanding voltage (at sea level):

(1-2) (5-6) 500 volts rms.
(3-4): 1000 volts rms.

Electrical characteristics:

No load: With 115 volts rms, 400 Hz across (1-2), current in (1-2) shall not exceed 90 milliamperes rms maximum and the power in (1-2) shall not exceed 2.0 watts.

Voltage across (3-4): 96.77 volts ±2 percent.

Voltage across (5-6): 5.26 volts ±2 percent

Rated load With 115 volts rms, 400 Hz across (1-2), the voltage across (3-4) shall be 92 0 volts rms ±3 percent at 20 milliamperes rms, voltage across (5-6) shall be 5.0 volts rms ±3 percent at 5.0 amperes rms

Polarity Additive, with terminals 2 and 5, 6 and 3 connected.

Temperature rise +40°C maximum with 115 volts rms, 400 Hz across (1-2) at ambient temperature of 110°C.

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DC resistance:

(1-2): 7.0 ohms maximum.
(3-4). 92.0 ohms maximum.
(5-6): 0.03 ohm maximum.

Regulation 6.0 percent maximum.

Ⓐ Quality assurance provisions:

Qualification inspection. Not applicable.

Quality conformance inspection: Groups A and B tests of MIL-T-27 shall be applicable.

Marking location: See figure 1.

Part number: M27/100-01.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 17, 80
DLA - ES

User activities:
Army - AR, ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0753-31)

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